

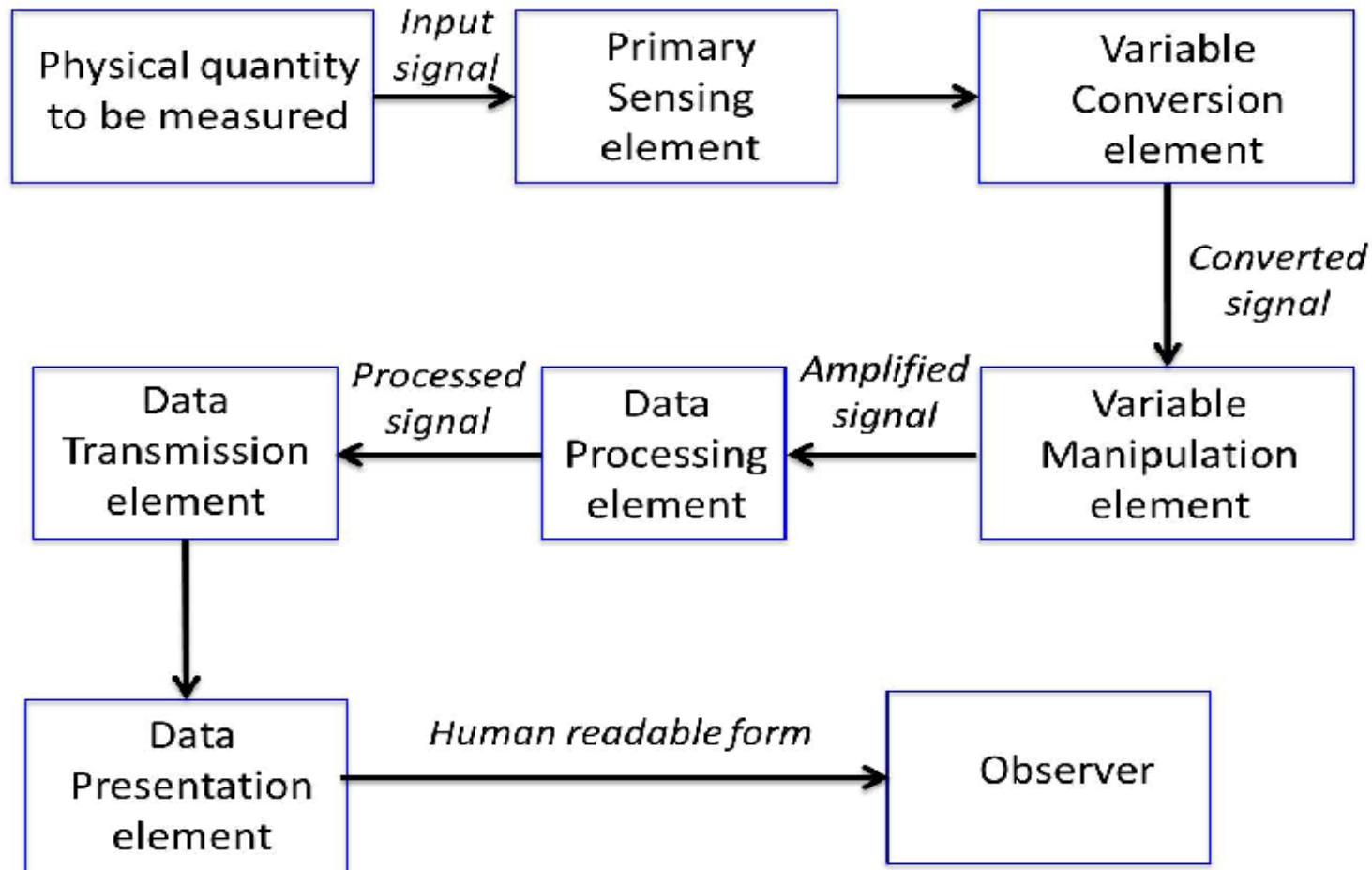
Applications of measuring instruments

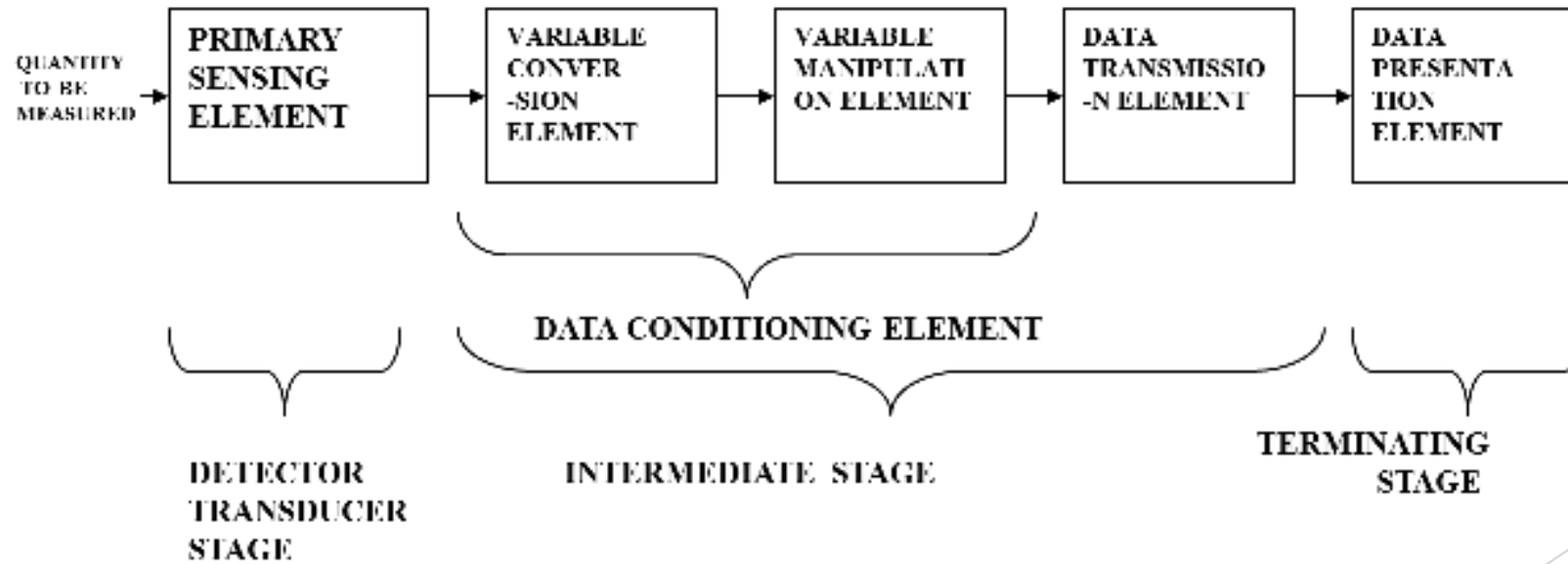
Generalized Measurement System

A generalized measurement system consists of the following components:

- ✧ 1. Primary Sensing Element
- ✧ 2. Variable Conversion Element
- ✧ 3. Variable Manipulation Element
- ✧ 4. Data Transmission System
- ✧ 5. Data Processing Element
- ✧ 6. Data Presentation Element

Generalized Measurement System





Functional Elements of Pressure Gauge

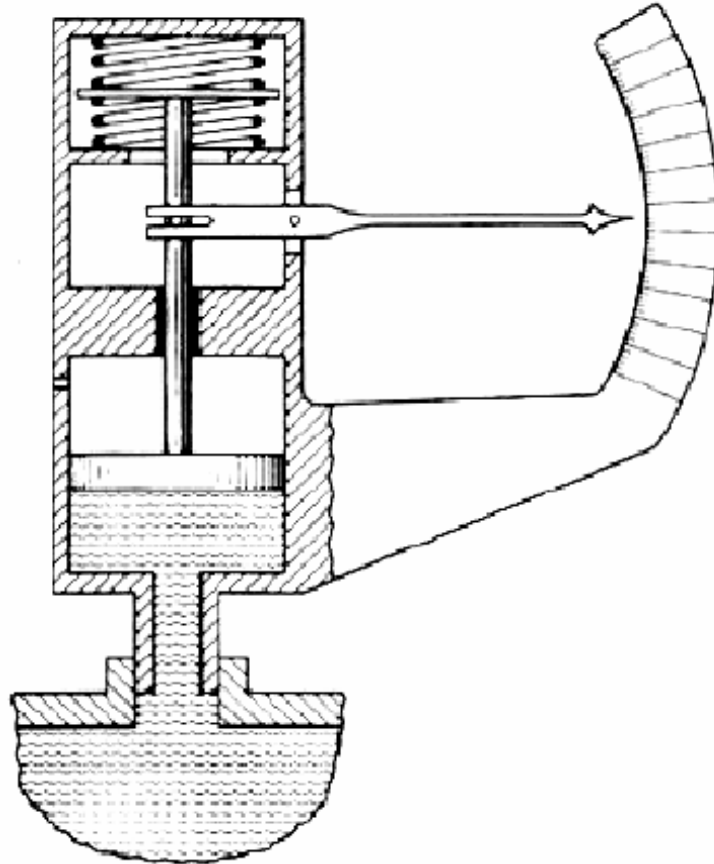
1) PISTON

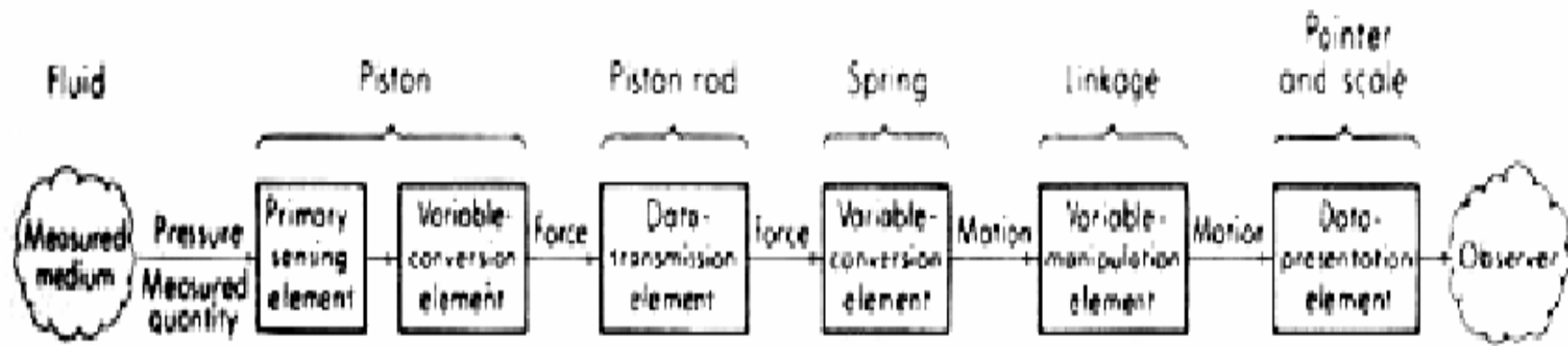
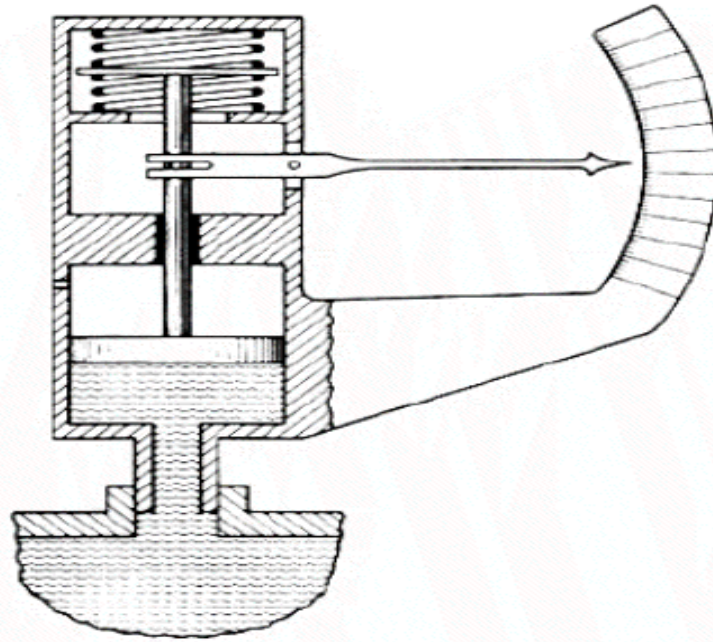
2) PISTON ROD

3) SPRING

4) LINKAGE

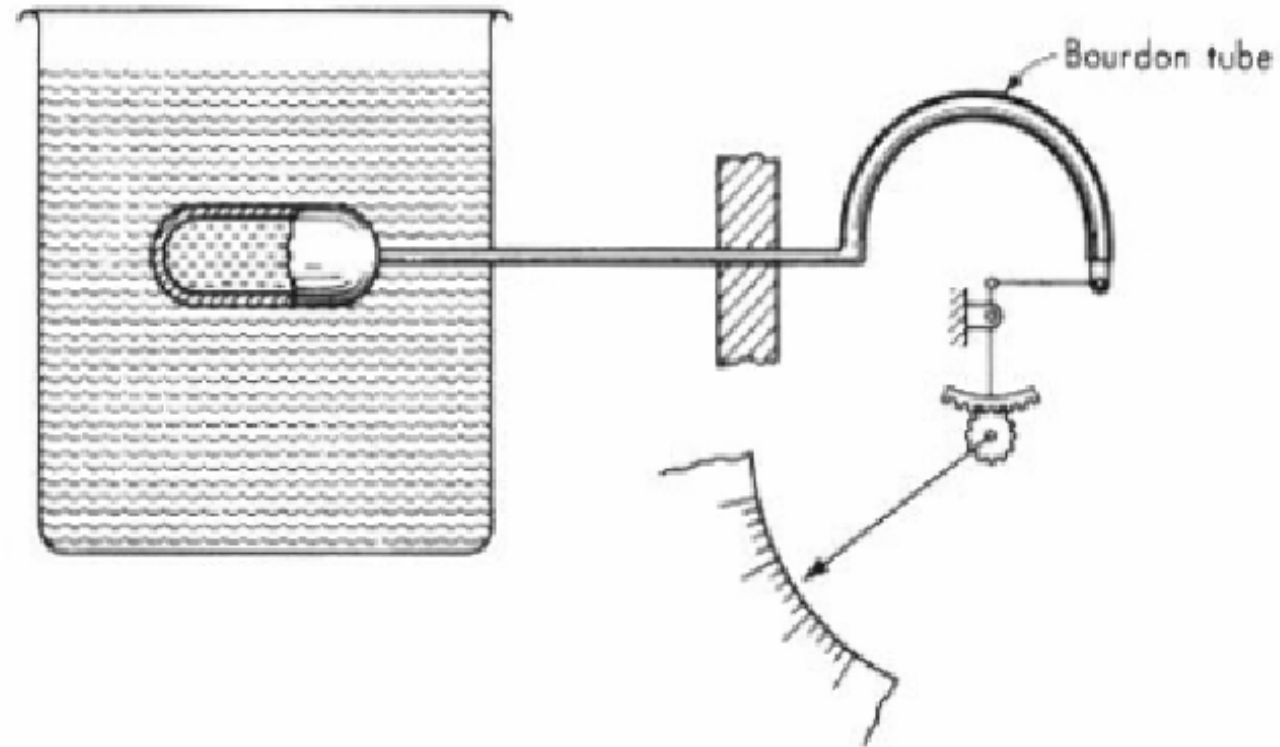
5) POINTER AND
SCALE

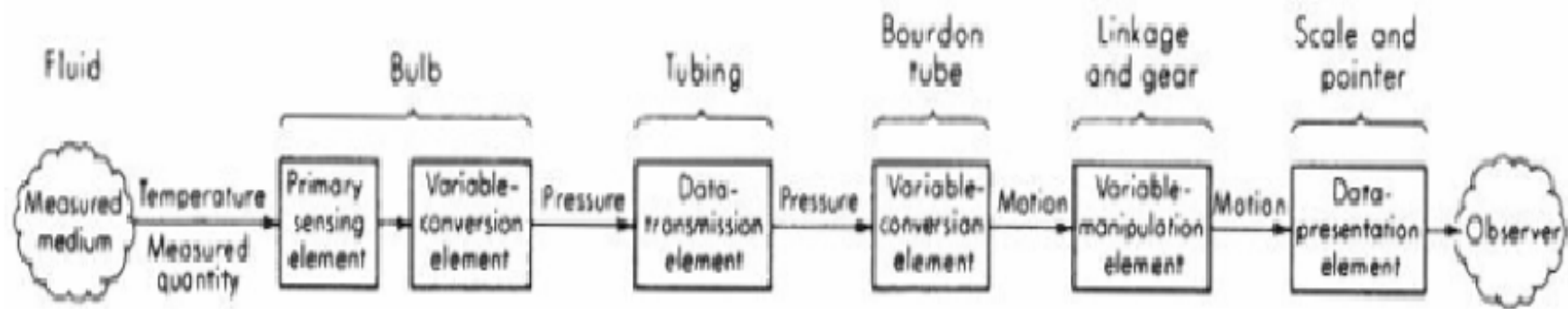
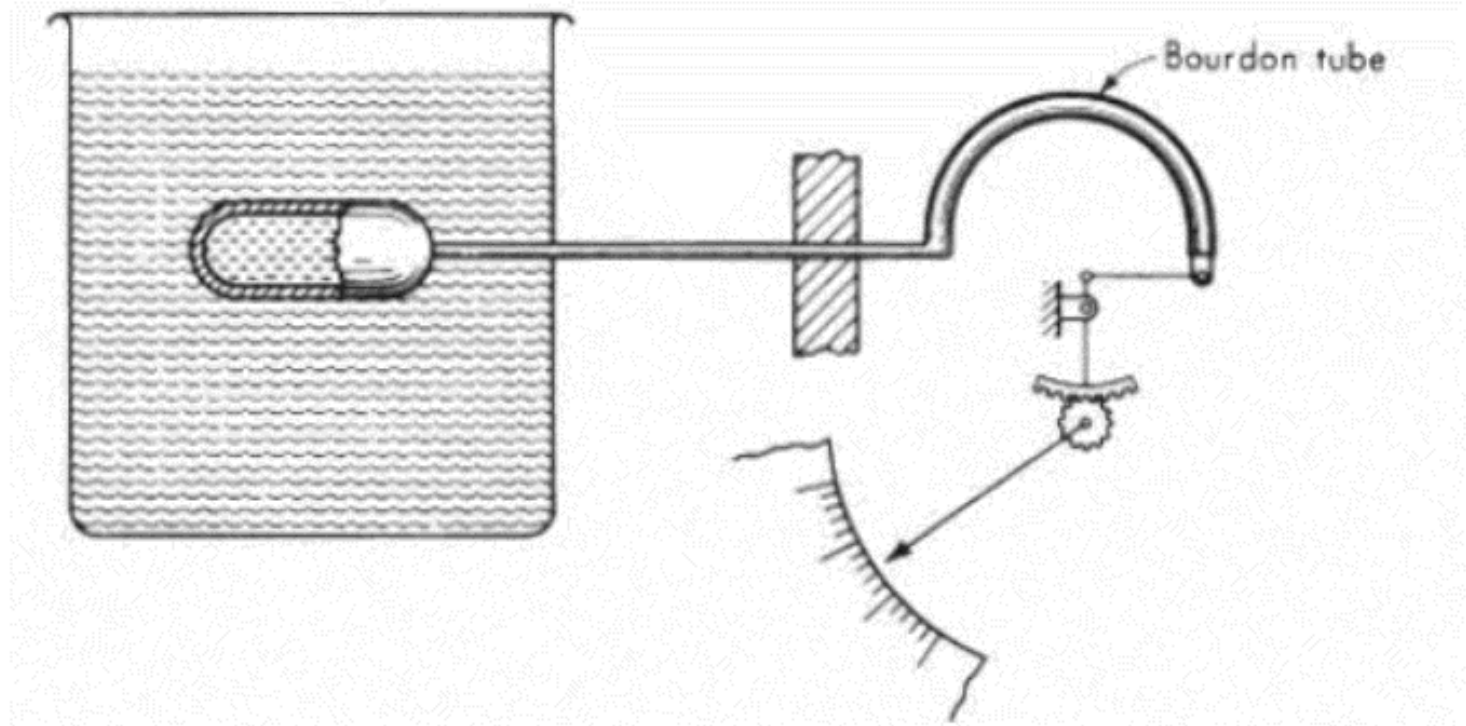




Functional Elements of Pressure Thermometer

- 1) BULB
- 2) TUBING
- 3) BURDON TUBE
- 4) LINKAGE AND GEAR
- 5) SCALE AND POINTER





Transducers

Transducers

- ⌘ A transducer is a device that converts energy from one form to another.
- ⌘ Energy forms can be mechanical, visual, electrical, thermal, chemical, etc



Transducers- classification

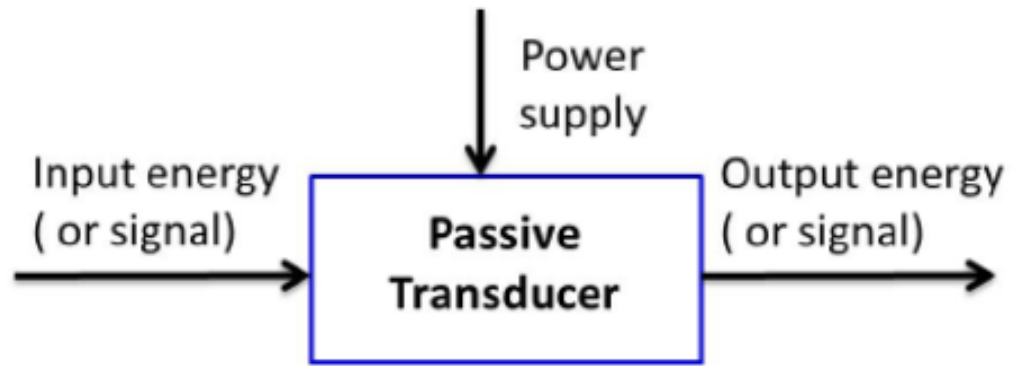
- ⊠ Transducers may be classified based on source of energy into two types,
 - Passive transducers and
 - Active transducers

Passive transducer

- ⊠ Passive transducers derive the power required for transduction from an auxiliary power source. They also derive part of power required for conversion from the physical quantity under measurement.

Active Transducers

- ⌘ An active transducer is one which does not require an auxiliary power source to produce their output. They are also known as self generating type since they develop their own voltage or current.



(a) Passive transducer



(b) Active transducer

Characteristics of measurement systems

Characteristics of measurement systems

- ⊠ The system characteristics are to be known, to choose an instrument that most suited to a particular measurement application
- ⊠ The performance characteristics may be broadly divided into two groups, namely
 - Static characteristics and
 - Dynamic characteristics

Static characteristics

- ⌘ The performance criteria for the measurement of quantities that remain constant, or vary only quite slowly.

OR

- ⌘ The set of criteria defined for the instrument which are used to measure the quantities that are varying slowly with time or constant is called static characteristics.

OR

- ⌘ The static characteristics of an instrument are considered for instruments which are used to measure an unvarying process condition.

Static characteristics

HOW?

- ⊠ All the static performance characteristics are obtained by one form or another of a process called **calibration**.

WHY?

- ⊠ It provides a opportunity to check the instrument against a known standard and to find the **errors and accuracy**.

Static characteristics of instruments

- ⊠ Accuracy & Precision
- ⊠ Sensitivity
- ⊠ Linearity & Hysteresis
- ⊠ Repeatability and Reproducibility
- ⊠ Resolution,
- ⊠ Drift,
- ⊠ Span
- ⊠ Threshold etc.

Static characteristics

- ⊠ **Accuracy:** The degree of exactness (closeness) of a measurement compared to the expected (true) value. It is expressed in terms of errors

$$\text{Static error} = \text{measured value} - \text{true value}$$

- ⊠ **Precision:** A measure of the consistency or reproducibility of measurements, i.e. successive readings does not differ.

Static characteristics

- ⊠ **Resolution:** It is the ability of the measurement system to detect and faithfully indicate small changes in the characteristic of the measurement result.
- ⊠ **Sensitivity:** The ratio of the change in output (response) of the instrument to a change of input.

Static characteristics

- ⊠ **Drift** : Gradual shift in the measured value ,over an extended period, when there is no change in input.
- ⊠ **Threshold**: The minimum value of input for which the device just starts to respond
- ⊠ **Range/Span**: The minimum and maximum value of quantity so that the device is capable of measuring

Static characteristics

- ⊠ **Repeatability:** A measure of how well the output returns to a given value when the same precise input is applied several times.

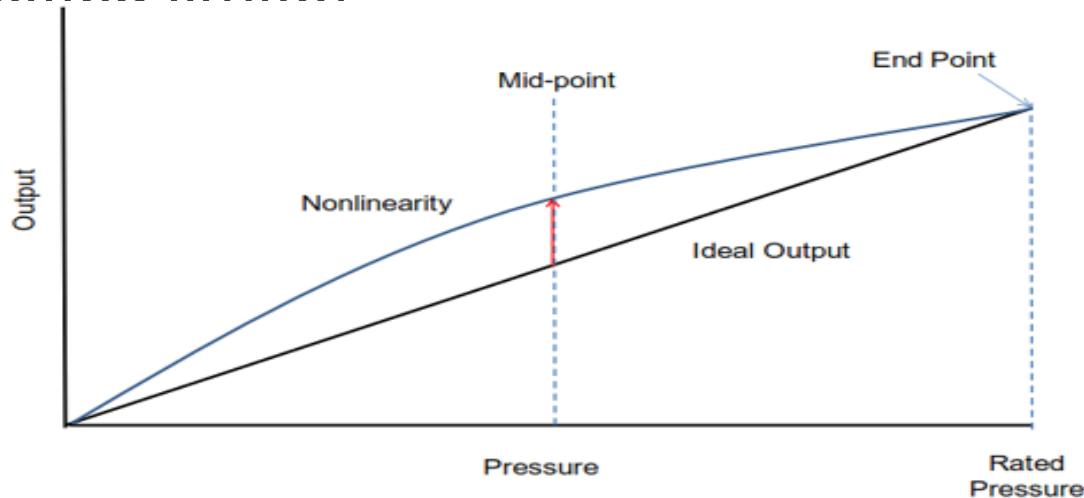
OR

- ⊠ The ability of an instrument to reproduce a certain set of reading within a given accuracy.

Static characteristics

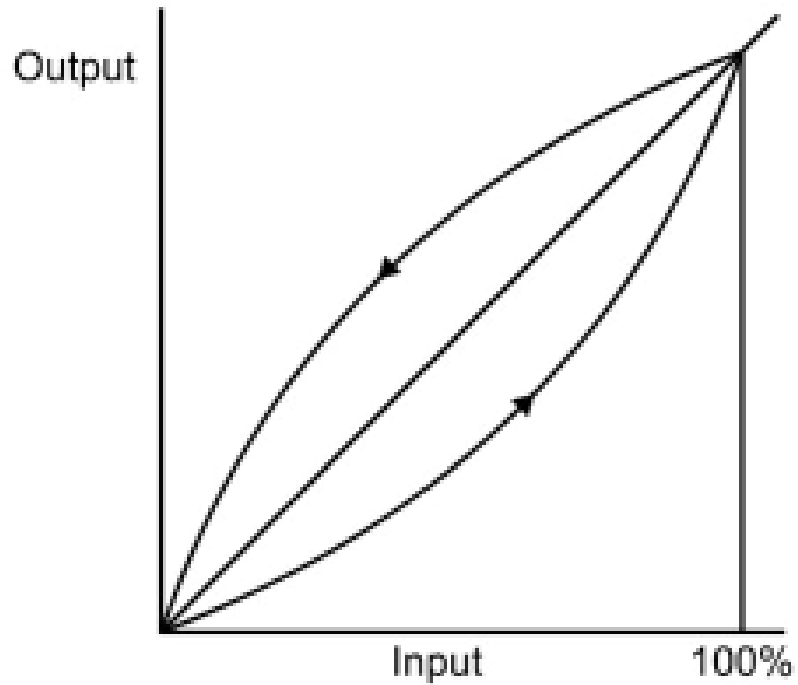
Linearity

- ⊠ Input output relationship of a device must be linear i.e, $Y = mx + C$
- ⊠ But practical systems shows small deviations from the linear shape (allowed within the specified limits)



Static characteristics

⊠ Hysteresis



Dynamic characteristics

Dynamic characteristics

- ⊠ The relationship between the system input and output when the measured quantity (measurand) is varying rapidly.
- ⊠ The dynamic characteristics of an instrument are
 - Speed of response
 - Fidelity
 - Time delay or lag
 - Dynamic error

Dynamic characteristics

- ⊠ **Speed of Response:** It is the ability of a system to respond to a sudden changes in the input signal/ quantity
- ⊠ **Fidelity:** It is the degree to which an instrument indicates the changes in the measured variable without dynamic error. (Indication of how much faithfully system responds to the changes in input).

Dynamic characteristics

- ⊠ **Lag:** It is the retardation or delay in the response of an instrument to changes in the measured variable.

Generalized mathematical model of instruments

Generalized Model of Instruments

- An ordinary differential equation of n^{th} order with constant coefficients can be considered to be a generalized model for an instrument
- Solution of this equation for known input will yield dynamic response

q_0 = output quantity

q_{in} = input quantity

a, b = constant coefficients, combination of system parameters

$$a_n \frac{d^n q_0}{dt^n} + a_{n-1} \frac{d^{n-1} q_0}{dt} + \dots + a_1 \frac{dq_0}{dt} + a_0 q_0 =$$
$$b_m \frac{d^m q_{in}}{dt^m} + b_{m-1} \frac{d^{m-1} q_{in}}{dt^{m-1}} + \dots + b_1 \frac{dq_{in}}{dt} + b_0 q_{in}$$

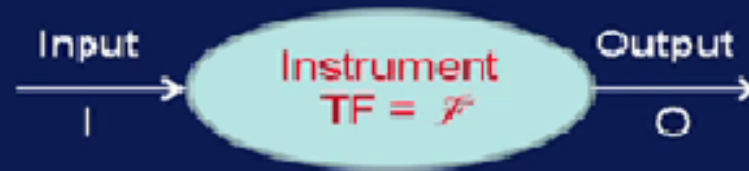
Generalized mathematical model of instruments

$$a_n \frac{d^n q_0}{dt^n} + a_{n-1} \frac{d^{n-1} q_0}{dt} + \dots + a_1 \frac{dq_0}{dt} + a_0 q_0 = b_0 q_{in}$$

- ⊠ Zero order instrument
- ⊠ First order instrument
- ⊠ Second order instrument

Input-Output Configuration of Instruments

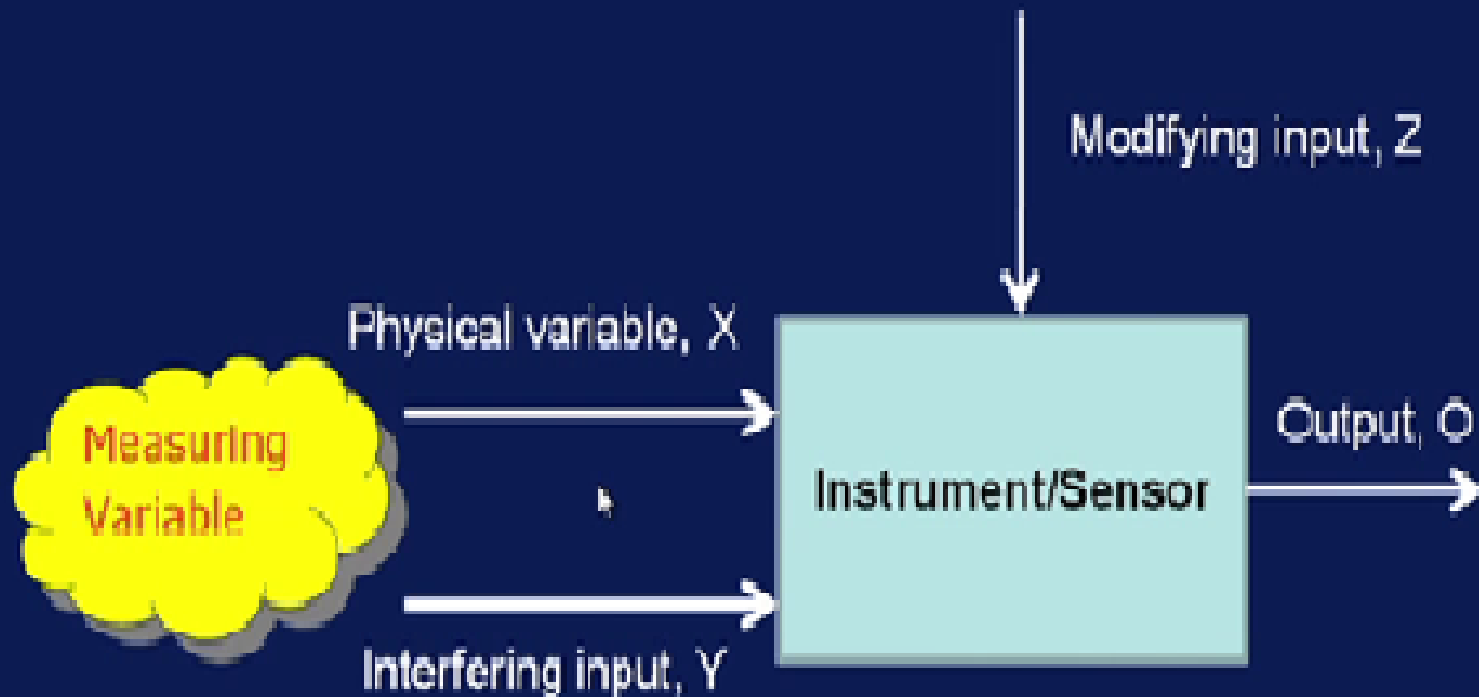
Can we develop a generalized configuration that represent significant input-output relationships present in an instrument?



Input quantities can be broadly classified into three categories:

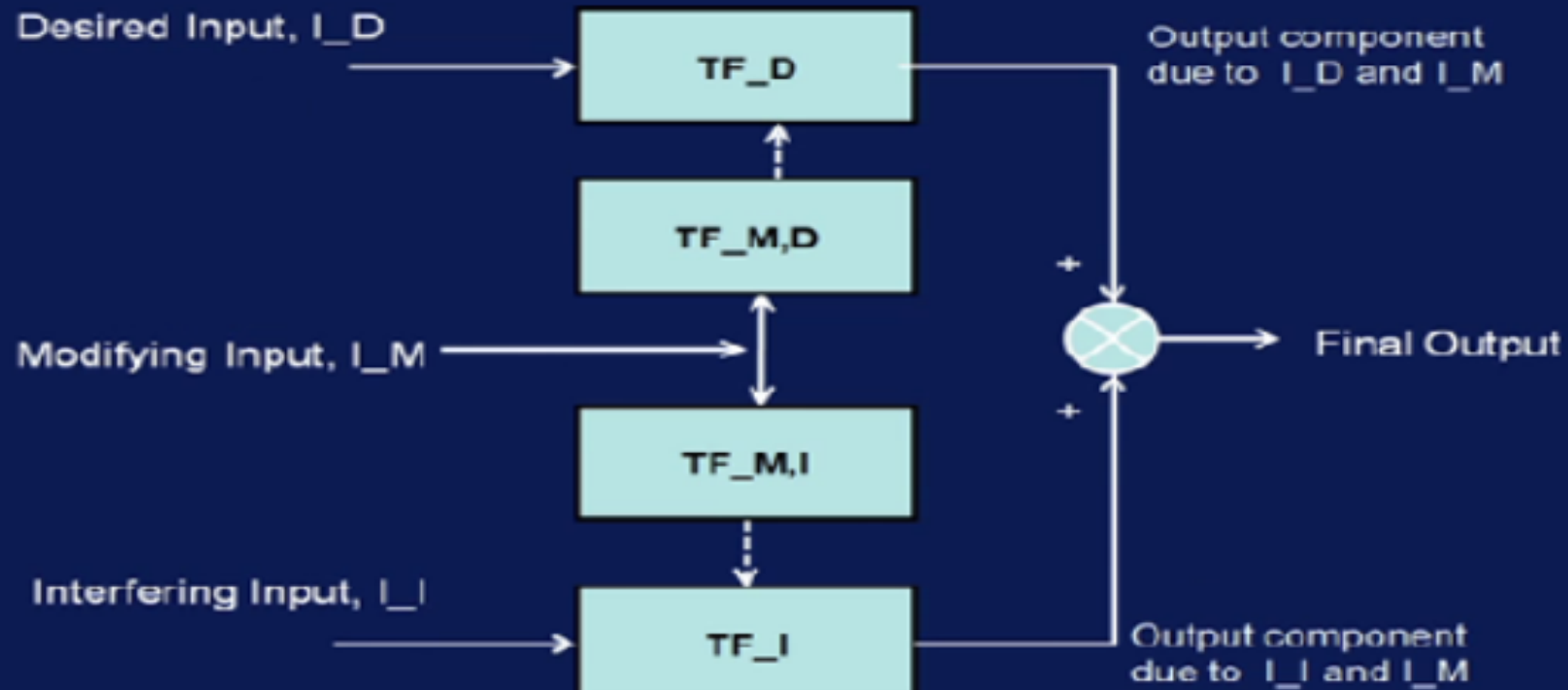
- **Desired Inputs:** measurands or quantities that the instrument is designed to be measured
- **Interfering Inputs:** quantities that unintentionally affect the instrument as a consequence of the principles used to acquire and process the desired inputs
- **Modifying Inputs:** undesired quantities that affect the output by altering the input-output relations for desired and interfering inputs

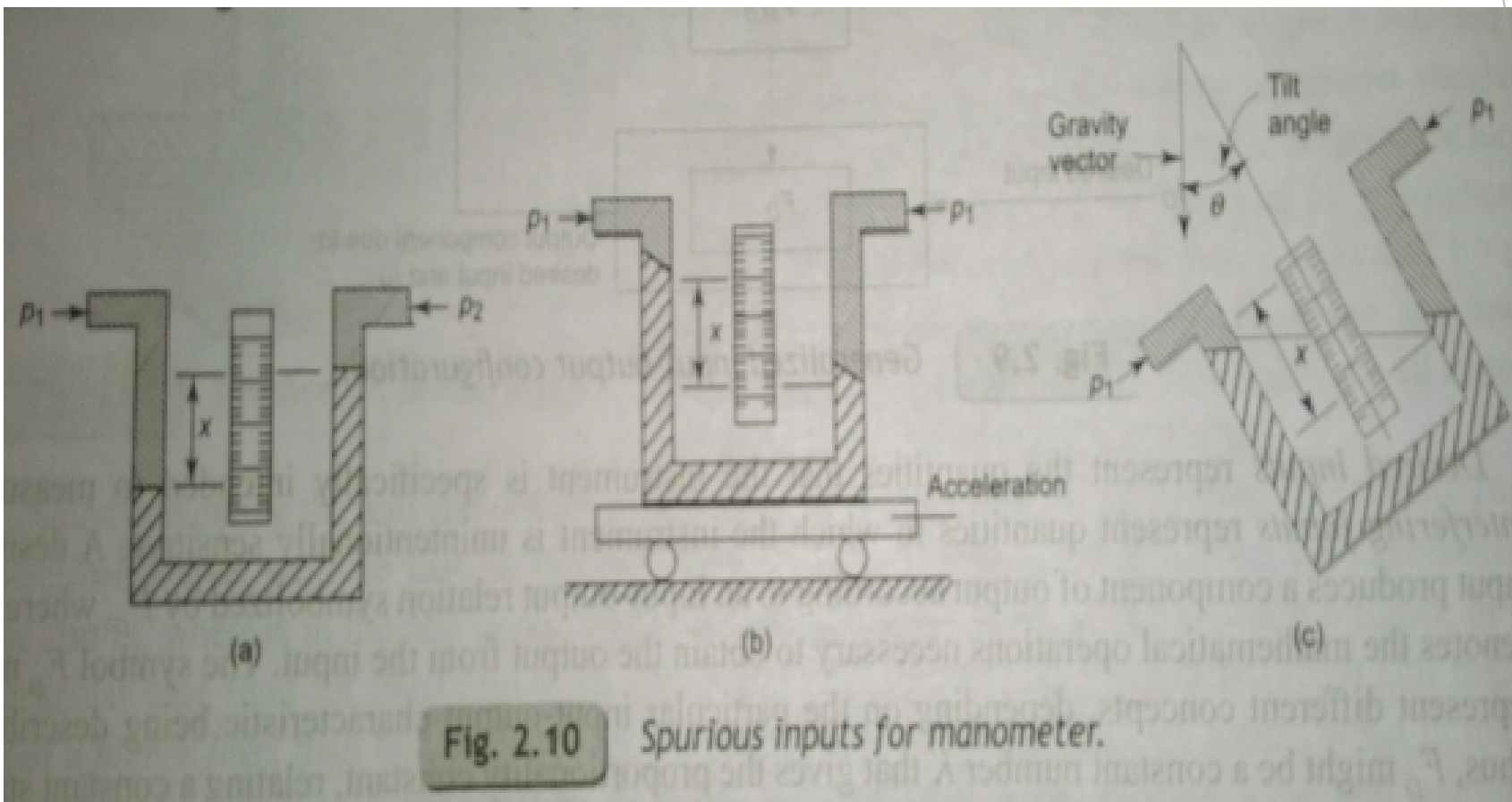
Input-Output Configuration of Instruments



Generalized Input-output configuration of measurement system

Input-Output Configuration of Instruments





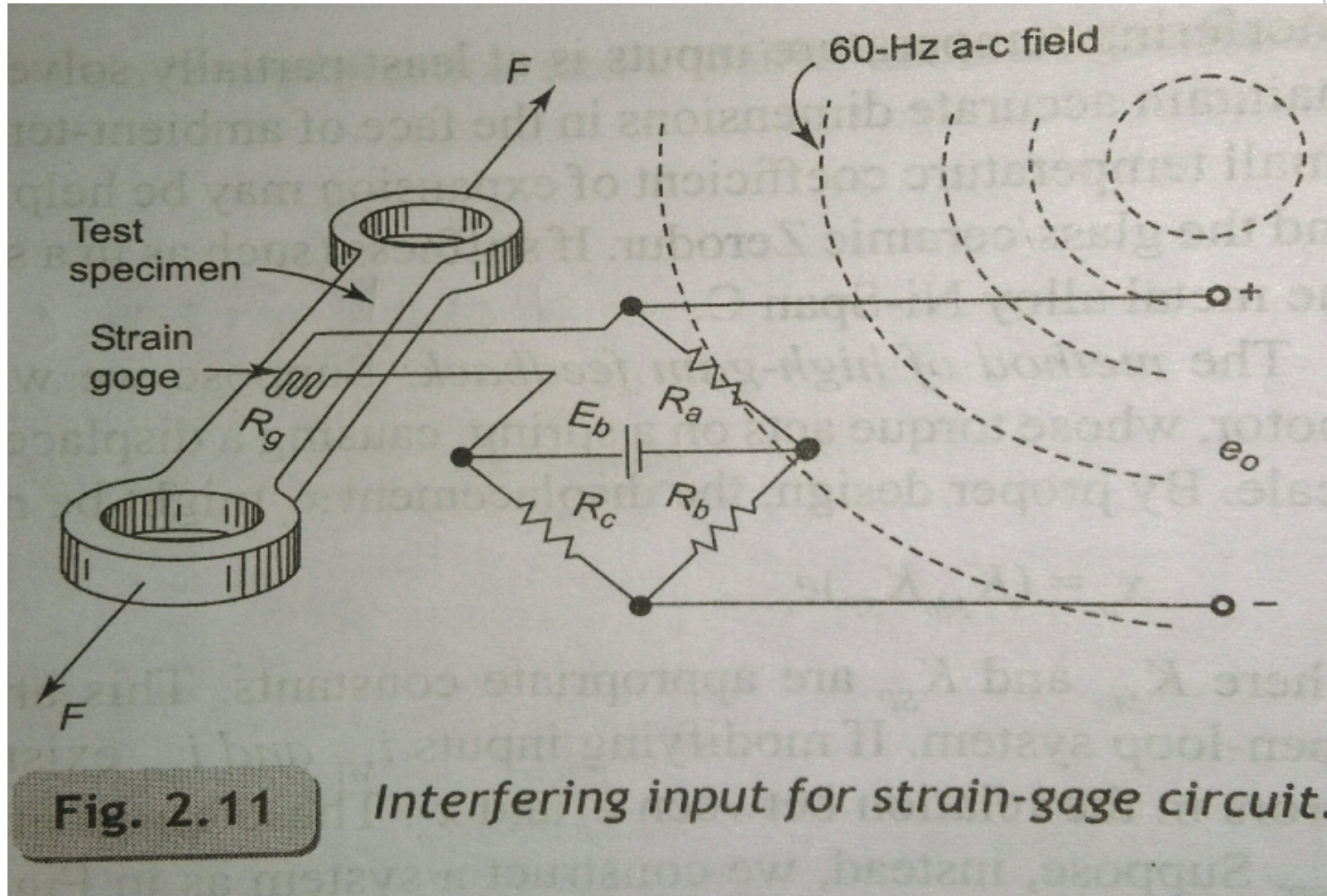


Fig. 2.11

Interfering input for strain-gage circuit.

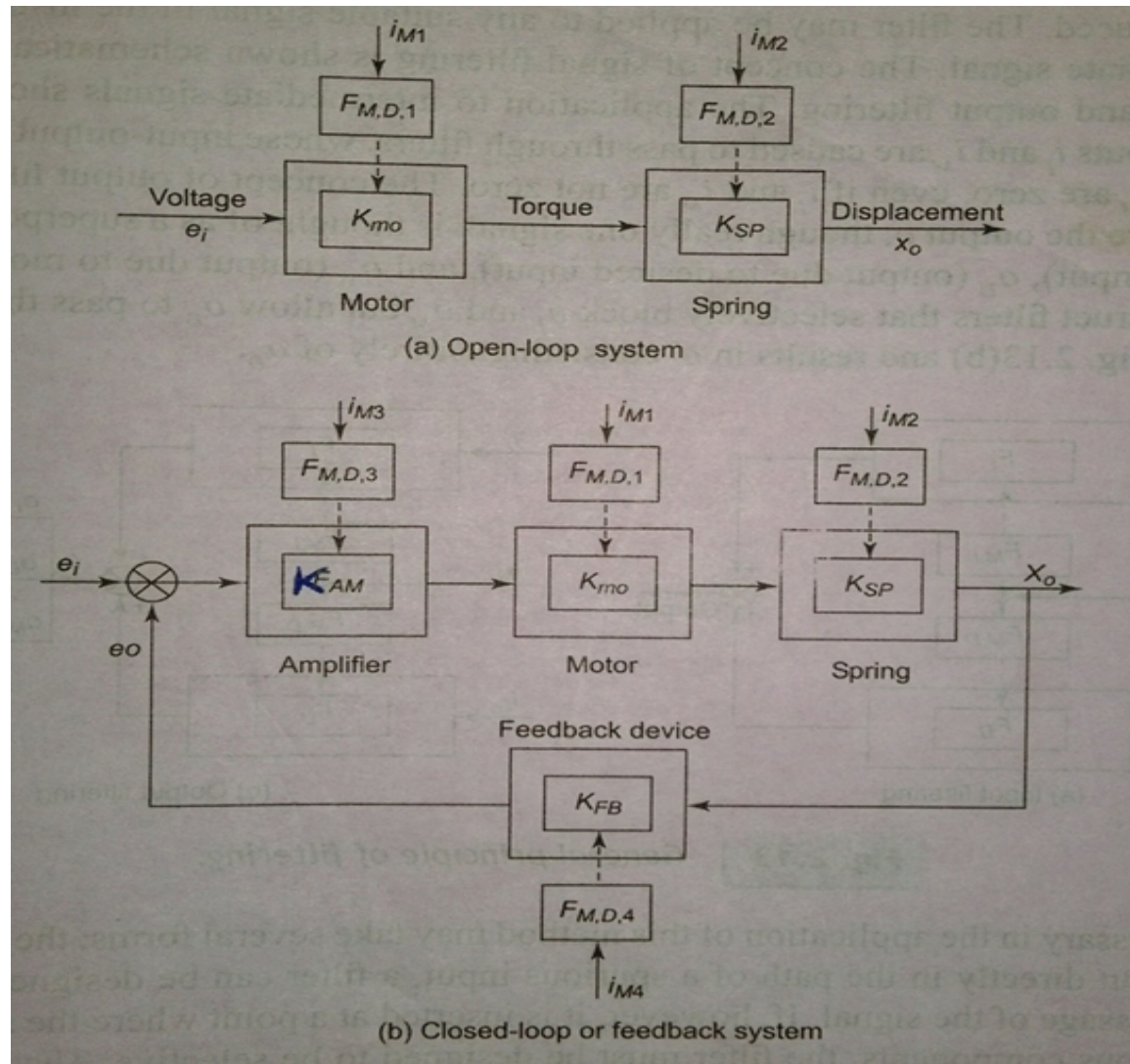
Methods Of Correction For Interfering And Modifying Input

1. Method of inherent insensitivity
2. Method of high gain feed back
3. Method of calculated output corrections
4. Method of signal filtering
5. Method of opposing input

1) Method of inherent insensitivity

- ⊠ sound design- sensitive to desired inputs only
- ⊠ In strain gauge- use a gauge material of 'low temperature coefficient of resistance'
- ⊠ In mech apparatus- use 'low temperature coefficient of expansion' materials..... Eg: metal alloy invar, glass ceramic zerodur.
- ⊠ If stiffness(in case of spring) must be temperature insensitive, consider nickel- iron- chromium alloy

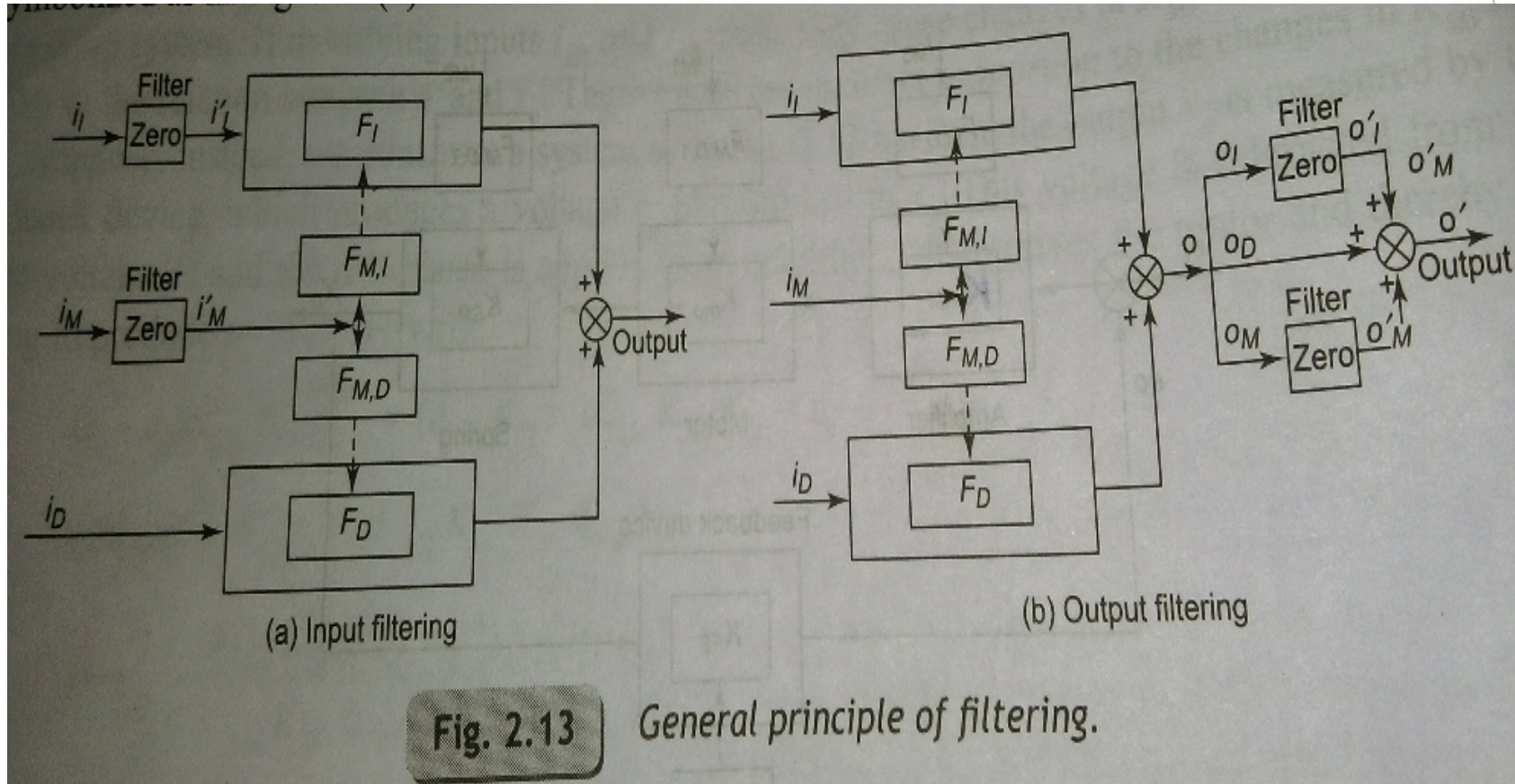
2) Method of high gain feed back



3) Method of calculated output corrections

- ⊠ Estimate the magnitudes of the **interfering and modifying inputs** and to know quantitatively how they affect the output
- ⊠ So it is possible to calculate corrections, which may be added or subtracted from the indicated output
- ⊠ In U-tube manometer example, the effect of temperature on both the calibrated scale's length and the density of mercury is found out for making suitable corrections in output

4) Method of signal filtering



5) Method of opposing input

